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King County Housing EDA — Client Presentation

Client & goal

- **Client:** ***Jennifer Montgomery*** (home buyer)
- **Goal:** Identify where "luxury" value shows up in King County using grade, condition, geographical data, price and size
- **Dataset:** King County house sales (features + sale price)

Project overview

- **What we did:** Exploratory Data Analysis (EDA) on house attributes vs. price
- **Main question:** Which combinations of features reliably predict the highest prices?
- **How to read this deck:** Each hypothesis shows (1) the idea, (2) the plot(s), (3) what it means for Jennifer

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Dataset at a glance

- **Target variable:** 💹Price, ↗️Grades, 💧Waterfront, 👌Condition
- **Key features used in hypotheses:**
    - Grade (overall construction and design quality)
    - Condition (overall state of the property)
    - Geographical location📍
    - Living area (sqft living)
    - Price (💰)
- *Other datasets which are considered but not necessarily used in understanding the data (Riverfront, Year renovated, Year built).*

Hypothesis 1 — *Grade & Condition Premium*

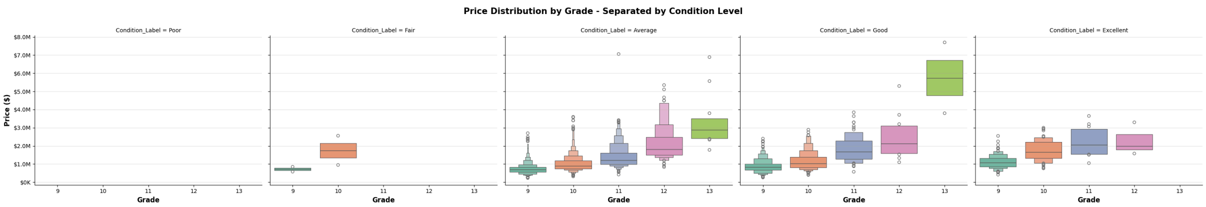
- **Claim:** *High grade (**10–13**) + good condition (**4–5**) produces the **highest prices***

  Broader market availability

- **Also true:** These homes are available across more of the county, not only in one area

EDA for hypothesis 1:

- Conditions by Grade × Price



Hypothesis 1 — *Findings*

- **Result:**
  - We found that better houses are in condition range Good - Excellent and grade higher than 10 are premium.
  - We can target these areas as per the Client's requirements.
- **Client takeaway:** Jennifer can search broadly, but should prioritize **grade 10+** and **condition 4–5**
- **Recommendation:** Use grade and condition as "must-have" filters before location fine-tuning

Hypothesis 2 — Size + Grade + Locality= Premium

- **Claim:** Large homes (4**000+ sqft**) with high grade command **premium prices**
- **Why it matters:** Spacious + high quality aligns with "luxury appeal" and resale demand

EDA for hypothesis 2:

- Price vs. sqft living (scattered with grade)

- Price distribution for 4000+ sqft vs. below

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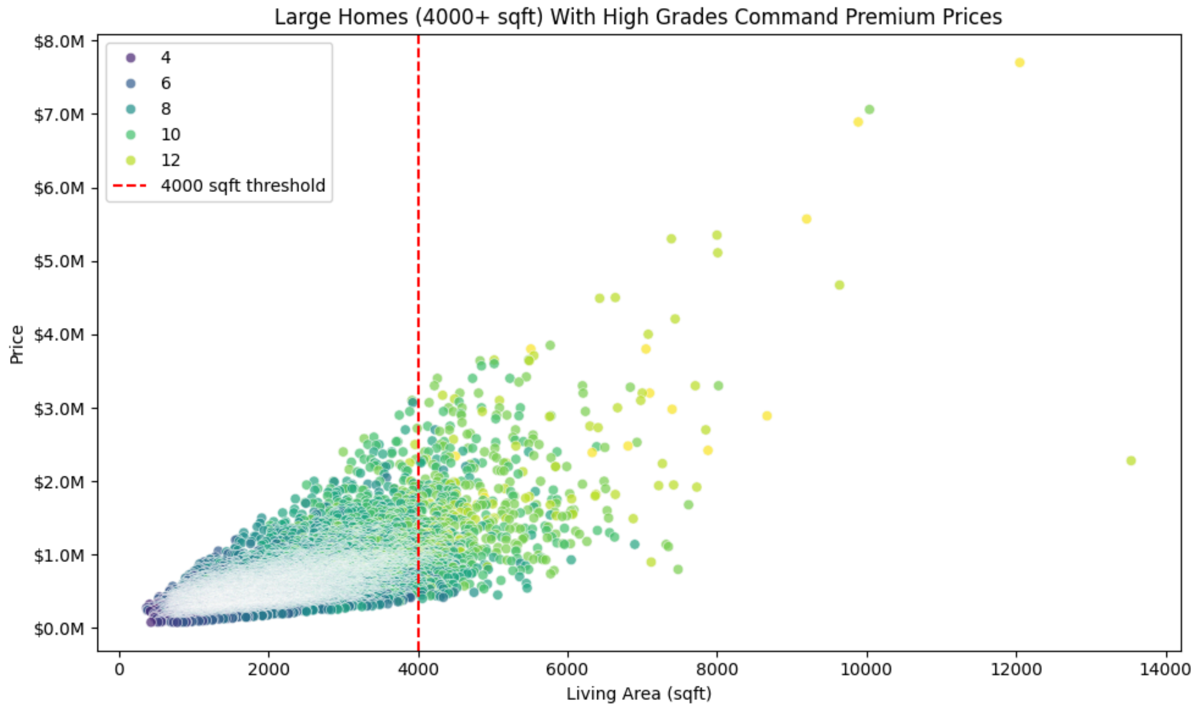
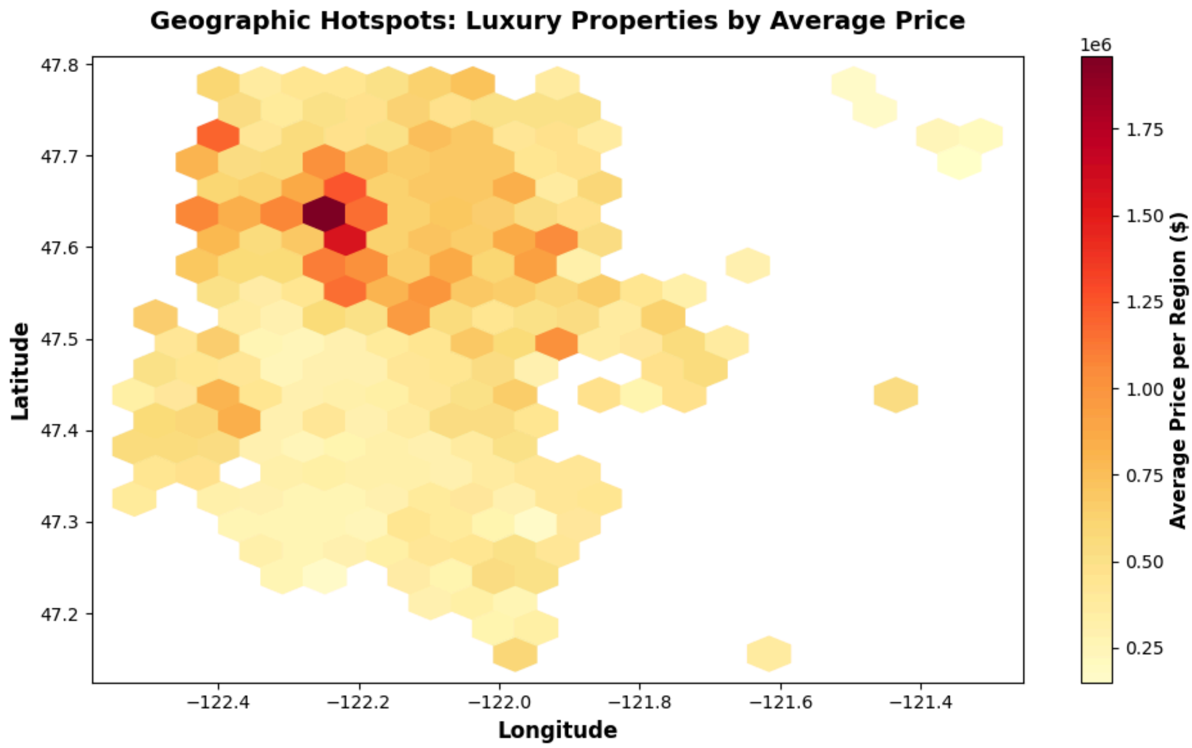


- Price vs Geographical data (Premium locations)



Hypothesis 2 — Findings

- **Result:**
  - *Higher grade observed above 4000 sqft living space with higher price range.*
  - *Good locality found between Lat > 47.5 and Long < -122.0*
- **Client takeaway:** For standout properties, size helps, but size alone is not enough without high grade and good location
- **Recommendation:** Shortlist homes that meet 4**000+ sqft** and **grade 10+** first, then assess location and features

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Hypothesis 3 — "Luxury Package" (Multi-factor)

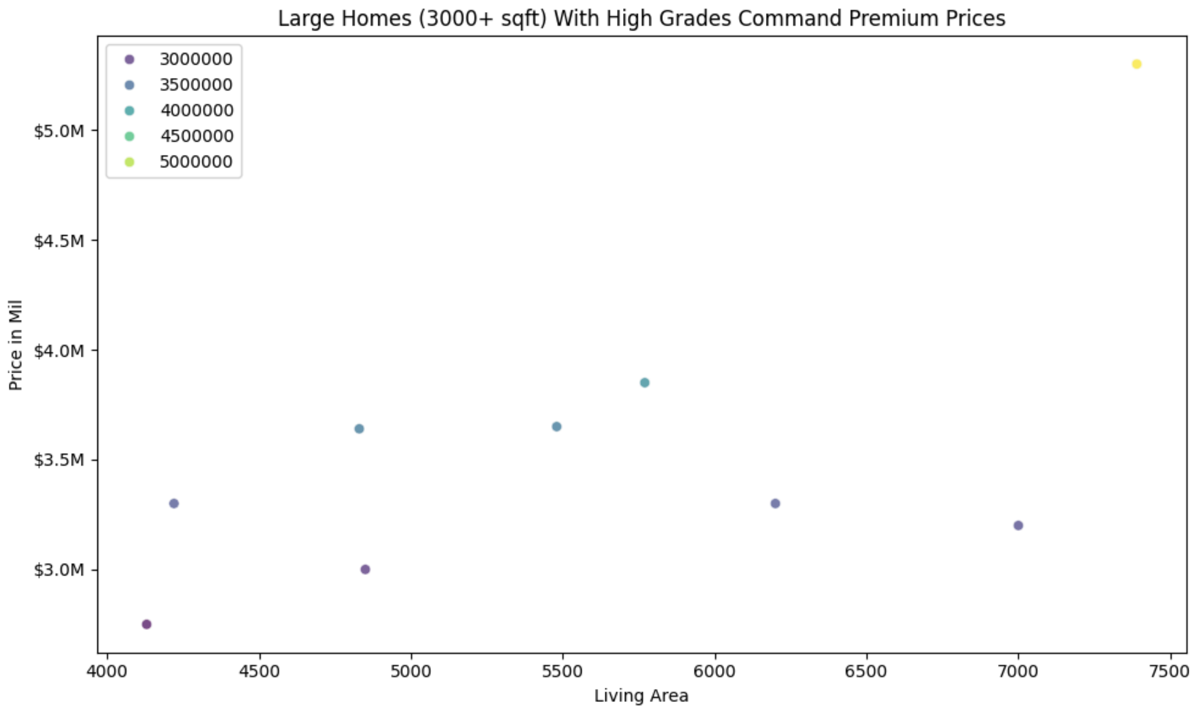
- **Definition:**
  - 💹Grade **10+**
  - 👌Condition **4+**
  - ⬜Size 4**000+ sqft**
  - 📍Latitude > 47.5 & Longitude < -122.0
  - 💧Waterfront = Yes

- **Claim:** This combination represents the **true luxury segment** and the hottest resale items

EDA for hypothesis 3:

- Distribution of prices for Luxury Package

- Count of homes meeting the package criteria



Hypothesis 3 — Findings

- **Luxury segment summary:**
  - 💰**Average price:** ***$3.55M***
  - **Typical range (IQR):** *$2.75M–$5.3M*
  - **Availability:** 9 homes

- **Client takeaway:** If Jennifer wants "true luxury," this package is the clearest filter and supports confident resale expectations

House ID	**Price** 💰
7558700030	$5.3M
8106100105	$3.85M
4217402115	$3.65M

2425049063	$3.64M
3625059043	$3.3M
3625059152	$3.3M
624069108	$3.2M
3761100045	$3.0M
624069035	$52.75M

Practical buying guidance (how to use this)

- **Step 1:** Decide if “Luxury Package” is a hard requirement or a nice-to-have
- **Step 2:** If budget is tight, relax one constraint (usually size first) and keep **grade** as the anchor
- **Step 3:** Verify with comps and neighborhood context for finalist homes